

Introduction

The study of cold neutral atoms continues to drive advancements in quantum simulation, precision metrology, and quantum information processing. The experiment described in this thesis, conducted at the cold atoms laboratory of the Department of Physics and Astronomy "A. Righi" (University of Bologna), aims to study a quantum system of cold rubidium-87 (^{87}Rb) gas confined inside a Hollow-Core Photonic Crystal Fiber (HCPCF). The integration of laser-cooled neutral atoms with hollow-core optical fibers allows for extended light-matter interactions compared with conventional free-space optical dipole traps. In free space, the interaction length is constrained by diffraction and the Rayleigh range. Hollow-core fibers bypass this limitation by confining atoms within a guided optical mode [1], maintaining a constant transverse profile over macroscopic distances. This geometry is relevant for improving the interrogation times of atomic magnetometers and matter-wave interferometers [2], as well as for exploring collective emission phenomena [3] and implementing quantum memories [1].

Loading cold atoms into the HCPCF requires an intermediate transport mechanism. The experimental sequence starts with a Magneto-Optical Trap (MOT) and an optical molasses phase to prepare a sub-Doppler atomic sample. The atoms are then loaded into an optical conveyor belt, which is formed by two counter-propagating Optical Dipole Trap (ODT) beams creating a dynamic standing wave.

This thesis describes the optimization of the atomic cloud preparation, the implementation of the second optical branch needed to complete the conveyor belt, and the numerical modeling of the Electromagnetically Induced Transparency (EIT) cooling required to reduce the atomic temperature once inside the fiber.

The manuscript is structured as follows. Chapter 1 provides the theoretical background on HCPCFs, optical trapping, conveyor belt kinematics, and EIT cooling. Chapter 2 describes the experimental apparatus, the imaging calibration, the design of the "black fiber" injection setup, and the formulation of the multi-level numerical model. Chapter 3 presents the data analysis and the experimental results, including the optimization of the molasses temperature and ODT capture efficiency, the power and polarization stability of the optical lattice, and the numerical simulations of the EIT cooling dynamics.